

Summer Math Booklet – Complete Answer Key

Grade 4→5 Transition · Peoria, AZ · iReady Aligned · Beast Academy Style

FOR PARENT/GUARDIAN USE: This answer key includes step-by-step solutions. The goal is for your child to learn FROM his mistakes, not just see the answers. After each day's work, have him check his own work using this key, explain any step he doesn't understand, and try a similar problem on his own.

How to Use This Key

1. Child completes the day's page independently.
2. Child checks answers using the key.
3. For wrong answers: child reads the step-by-step solution slowly, then covers it and re-solves the problem.
4. If still confused: parent walks through the example together.
5. Quiz answers are given at the end — do not share these until the quiz is done.

Topics by Week:

- W1** Number Sense — Place Value, Rounding, Comparing, Expanded Form
- W2** Fractions — Equivalence, Comparing, Adding/Subtracting, Mixed Numbers
- W3** Multiplication & Division — Multi-digit, Long Division, Factors, Word Problems
- W4** Decimals — Place Value, Reading/Writing, Comparing, All Operations
- W5** Geometry — Angles, Triangles, Quadrilaterals, Area/Perimeter, Coordinates
- W6** Exponents — Base/Exponent, Powers of 10, PEMDAS, Perfect Squares & $\sqrt{\quad}$
- W7** Expressions & Equations — Variables, Like Terms, 1-Step & 2-Step, Inequalities
- W8** Probability — P Formula, Scale, Experimental/Theoretical, Tree Diagrams, Predictions



Week 6–8 Answer Key Location:

Answers for Weeks 6, 7, and 8 are printed at the end of **summer_booklet_weeks6_8.html**.
Open that file and scroll to the last section titled "Answer Key – Weeks 6, 7 & 8".



Navigation Hub:

Open **index.html** in any browser to navigate all booklets, comic guides, and answer keys from one place. Share that URL with your son so he can access everything from his own computer.

Week 1 Answer Key – Number Sense & Operations

Day 1 – Place Value

1. 3,462,805

Place: **hundred-thousands**

Value: **400,000**

Step: Count from right: ones, tens, hundreds, thousands, ten-thousands, *hundred-thousands*. The 4 is in that place, so value
 $= 4 \times 100,000 = 400,000$.

2. 27,914,600

Place: **millions**

Value: **9,000,000**

3. 513,002,744

Place: **hundred-millions**

Value: **500,000,000**

4. 8,004,371

Place: **hundreds**

Value: **300**

5. 605,814

600,000 + 5,000 + 800 + 10 + 4

Step: Write the value of each non-zero digit. The 0 in the ten-thousands place contributes nothing, so skip it.

6. 3,070,009

3,000,000 + 70,000 + 9

7. $400,000 + 60,000 + 500 + 7 =$ **460,507**

8. $3,841,002 < 3,841,020$

*Same through ten-thousands. Ones: 2 vs
20 $\rightarrow$ less.*

9. $70,500,000 > 7,500,000$

8 digits vs 7 digits $\rightarrow$ larger.

10. $999,999 < 1,000,000$

6 digits vs 7 digits.

11. $24,060,000 > 24,006,000$

Same millions. Ten-thousands: 6 vs 0.

12. Round 4,738,201 to nearest million.

5,000,000

*Look at hundred-thousands: $7 \geq 5 \rightarrow$
round up.*

13. Round 583,647 to nearest 10,000.

580,000

*Look at thousands digit: $3 < 5 \rightarrow$ round
down.*

14. Round 29,450 to nearest thousand. **29,000**

Hundreds digit is $4 < 5 \rightarrow$ round down.

15. Round 7,499,999 to nearest million. **7,000,000**

*Hundred-thousands digit: $4 < 5 \rightarrow$ round down. Many students say
8 million – remember to look at the digit **JUST AFTER** the place
you're rounding to.*

Challenge: 3,006,009

Millions digit = 3. Thousands digit = $2 \times 3 = 6$. Ones digit = $3 + 6 = 9$. All others = 0.

$\rightarrow 3,000,000 + 6,000 + 9 = 3,006,009$

Day 2 – Rounding & Estimation

Part A – Round It Table:

4,763: nearest 10 = **4,760**, nearest 100 = **4,800**, nearest 1,000 = **5,000**

28,549: nearest 10 = **28,550**, nearest 100 = **28,500**, nearest 1,000 = **29,000**

305,281: nearest 10 = **305,280**, nearest 100 = **305,300**, nearest 1,000 = **305,000**

1,999,500: nearest 10 = **1,999,500**, nearest 100 = **1,999,500**, nearest 1,000 = **2,000,000**

Week 1 Answer Key (continued)

Day 4 – Multiplication

1. $83 \times 10 = \mathbf{830}$; $83 \times 100 = \mathbf{8,300}$;
 $83 \times 1,000 = \mathbf{83,000}$

Just add one zero, two zeros, three zeros.

2. $7 \times 40 = \mathbf{280}$; $7 \times 400 = \mathbf{2,800}$;
 $70 \times 400 = \mathbf{28,000}$

Distributive Property steps:

3. $6 \times 47 = 6 \times (40 + 7) = 240 + 42 = \mathbf{282}$

4. $8 \times 93 = 8 \times (90 + 3) = 720 + 24 = \mathbf{744}$

5. $5 \times 108 = 5 \times (100 + 8) = 500 + 40 = \mathbf{540}$

6. $12 \times 25 = 12 \times (20 + 5) = 240 + 60 = \mathbf{300}$

7. 346×7

$6 \times 7 = 42$ write 2 carry 4

$4 \times 7 = 28 + 4 = 32$ write 2 carry 3

$3 \times 7 = 21 + 3 = 24$

$= \mathbf{2,422}$

8. 524×38

$524 \times 8 = 4,192$

$524 \times 30 = 15,720$

Total = $\mathbf{19,912}$

9. $2,004 \times 6$

$4 \times 6 = 24$ write 4 carry 2

$0 \times 6 + 2 = 2$

$0 \times 6 = 0$

$2 \times 6 = 12$

$= \mathbf{12,024}$

10. Pencils per year:

Step 1: $48 \text{ boxes} \times 144 \text{ pencils} = 6,912 \text{ pencils/month}$

Step 2: $6,912 \times 9 \text{ months} = \mathbf{62,208 \text{ pencils}}$

Estimate: $50 \times 150 \times 9 \approx 67,500$ (close ✓)

Challenge A: $25 \times 16 = 400$. $50 \times 8 = 400$. **TRUE**. Reasoning: $25 \times 16 = 25 \times (2 \times 8) = (25 \times 2) \times 8 = 50 \times 8$ (Associative Property).

Challenge B: Method 1: $15 \times (20 + 4) = 300 + 60 = 360$. Method 2: $(10 + 5) \times 24 = 240 + 120 = 360$.

Day 5 – Division

1. $97 \div 4$

$4 \times 24 = 96$, remainder 1

Q = $\mathbf{24}$ R = $\mathbf{1}$

2. $253 \div 7$

$7 \times 36 = 252$, remainder 1

Q = $\mathbf{36}$ R = $\mathbf{1}$

3. $8,145 \div 9$

$9 \times 905 = 8,145$, remainder 0

Q = $\mathbf{905}$ R = $\mathbf{0}$

4. $75 \div 8 = 9 \text{ R } 3 \rightarrow$ Need $\mathbf{10 \text{ vans}}$. You must ROUND UP because 9 vans only carry 72 kids — 3 are left out. Always round up when you need to fit everyone.

5. $130 \div 12 = 10 \text{ R } 10 \rightarrow \mathbf{10 \text{ full boxes}}$, 10 leftover cookies. Round DOWN — you can only fill COMPLETE boxes.

6. $47 \div 4 = 11 \text{ R } 3$. Each friend gets $\mathbf{\$11}$. Leftover: $\mathbf{\$3}$.

7. $4,368 \div 12$

12 into 43 = 3 ($3 \times 12 = 36$), subtract $\rightarrow 7$; bring down 6 $\rightarrow 76$

12 into 76 = 6 ($6 \times 12 = 72$), subtract $\rightarrow 4$; bring down 8 $\rightarrow 48$

12 into 48 = 4. Answer: $\mathbf{364}$

Check: $364 \times 12 = 4,368$ ✓

8. $3,525 \div 15$

15 into 35 = 2, remainder 5; bring down 2 $\rightarrow 52$

15 into 52 = 3 ($3 \times 15 = 45$), remainder 7; bring down 5 $\rightarrow 75$

15 into 75 = 5. Answer: $\mathbf{235}$

Check: $235 \times 15 = 3,525$ ✓

Week 2 Answer Key – Fractions

Day 1 – Equivalence & Comparing

- $\frac{3}{4} = \frac{6}{8}$ (multiply top and bottom by 2). Shade 3 out of 4 sections.
- $\frac{2}{5} = \frac{6}{15}$ ($\times 3$)

$$3. \frac{3}{5} = \frac{12}{20} (\times 4)$$

$$4. \frac{4}{6} = \frac{2}{3} (\div 2)$$

$$5. \frac{5}{8} = \frac{25}{40} (\times 5)$$

$$6. \frac{18}{24} = \frac{3}{4} (\div 6 - \text{GCF of 18 and 24 is 6})$$

Simplest Form: 7. $\frac{8}{12} = \frac{2}{3}$ (GCF=4, divide both by 4) 8. $\frac{15}{20} = \frac{3}{4}$ (GCF=5) 9. $\frac{24}{36} = \frac{2}{3}$ (GCF=12) 10. $\frac{30}{45} = \frac{2}{3}$ (GCF=15)

Comparing (use LCD):

- $\frac{3}{4}$ vs $\frac{5}{8}$: LCD=8. $\frac{6}{8}$ vs $\frac{5}{8} \rightarrow >$
- $\frac{2}{3}$ vs $\frac{3}{5}$: LCD=15. $\frac{10}{15}$ vs $\frac{9}{15} \rightarrow >$
- $\frac{7}{8}$ vs $\frac{11}{12}$: LCD=24. $\frac{21}{24}$ vs $\frac{22}{24} \rightarrow <$
- $\frac{4}{6}$ vs $\frac{6}{9}$: Both = $\frac{2}{3} \rightarrow =$

Challenge – Order: LCD of 6,4,12,3 = 12.

$$\frac{5}{6} = \frac{10}{12}, \frac{3}{4} = \frac{9}{12}, \frac{7}{12} = \frac{7}{12}, \frac{2}{3} = \frac{8}{12}$$

$$\text{Order: } \frac{7}{12} < \frac{2}{3} < \frac{3}{4} < \frac{5}{6}$$

Day 2 – Adding & Subtracting Unlike Fractions

LCD: 1. 4&6→**12** 2. 3&8→**24** 3. 5&6→**30** 4. 4&9→**36**

- $\frac{1}{2} + \frac{1}{5}$: LCD=10. $\frac{5}{10} + \frac{2}{10} = \frac{7}{10}$
- $\frac{2}{3} + \frac{1}{4}$: LCD=12. $\frac{8}{12} + \frac{3}{12} = \frac{11}{12}$
- $\frac{3}{4} + \frac{2}{5}$: LCD=20. $\frac{15}{20} + \frac{8}{20} = \frac{23}{20} = 1 \frac{3}{20}$
- $\frac{5}{6} + \frac{3}{8}$: LCD=24. $\frac{20}{24} + \frac{9}{24} = \frac{29}{24} = 1 \frac{5}{24}$

- $\frac{3}{4} - \frac{1}{3}$: LCD=12. $\frac{9}{12} - \frac{4}{12} = \frac{5}{12}$
- $\frac{5}{6} - \frac{1}{4}$: LCD=12. $\frac{10}{12} - \frac{3}{12} = \frac{7}{12}$
- $\frac{7}{8} - \frac{2}{3}$: LCD=24. $\frac{21}{24} - \frac{16}{24} = \frac{5}{24}$
- $1 - \frac{3}{7}$: LCD=7. $\frac{7}{7} - \frac{3}{7} = \frac{4}{7}$

- $\frac{2}{3} + \frac{1}{4} = \frac{8}{12} + \frac{3}{12} = \frac{11}{12}$ of a bottle. Less than full ($\frac{12}{12}$).

- $\frac{3}{4} - \frac{1}{3} = \frac{9}{12} - \frac{4}{12} = \frac{5}{12}$ cup more needed.

Week 3 Answer Key – Multiplication, Division & Number Theory

Day 1 – Multi-Digit Multiplication

1. 73×46 : $73 \times 6 = 438$; $73 \times 40 = 2,920$; $438 + 2,920 = \mathbf{3,358}$
2. 258×37 : $258 \times 7 = 1,806$; $258 \times 30 = 7,740$; sum = $\mathbf{9,546}$
3. 614×83 : $614 \times 3 = 1,842$; $614 \times 80 = 49,120$; sum = $\mathbf{50,962}$
4. 307×59 : $307 \times 9 = 2,763$; $307 \times 50 = 15,350$; sum = $\mathbf{18,113}$
5. $4,025 \times 14$: $4,025 \times 4 = 16,100$; $4,025 \times 10 = 40,250$; sum = $\mathbf{56,350}$
6. 528×406 : $528 \times 6 = 3,168$; $528 \times 400 = 211,200$; sum = $\mathbf{214,368}$

7. Area model 63×47 : $(60 \times 40) = 2,400$ + $(60 \times 7) = 420$ + $(3 \times 40) = 120$ + $(3 \times 7) = 21$. Total = $\mathbf{2,961}$

8. Seats: $48 \times 36 = 1,728$. Revenue: $1,728 \times \$12 = \mathbf{\$20,736}$
9. $58 \times 24 = \mathbf{1,392 \text{ miles}}$. Rounded to nearest hundred: $\mathbf{1,400}$

Challenge: $4_ \times 37 = 1,7_6$. Try 4: $44 \times 37 = 1,628$ ✗. $46 \times 37 = 1,702$ ✗. $47 \times 37 = 1,739$ ✗. $48 \times 37 = 1,776$ ✓. Missing digit = $\mathbf{8}$. Full: $48 \times 37 = 1,776$.

Day 2 – Long Division

1. $5,376 \div 24 = \mathbf{224}$. Check: $224 \times 24 = 5,376$ ✓
Steps: 24 into 53 = 2 ($2 \times 24 = 48$), r=5; bring 7 $\rightarrow$ 57; 24 into 57 = 2 ($2 \times 24 = 48$), r=9; bring 6 $\rightarrow$ 96; 24 into 96 = 4 ($4 \times 24 = 96$), r=0.
2. $7,056 \div 48 = \mathbf{147}$. Check: $147 \times 48 = 7,056$ ✓
3. $4,891 \div 13$: Q = $\mathbf{376}$ R = $\mathbf{3}$. Check: $376 \times 13 + 3 = 4,891$ ✓
4. $9,072 \div 36 = \mathbf{252}$. Check: $252 \times 36 = 9,072$ ✓

5. $8,463 \div 27$: Est $8,000 \div 30 \approx 267$. Exact = $\mathbf{313 \text{ R } 12}$
6. $6,480 \div 32$: Est $6,400 \div 32 = 200$. Exact = $\mathbf{202 \text{ R } 16}$

7. $2,496 \div 32 = \mathbf{78 \text{ full boxes}}$
8. $5,040 \div 14 = \mathbf{360 \text{ students per room}}$. $360 \div 4 = \mathbf{90 \text{ tables per room}}$

Challenge: $148 \times 25 = 3,700$. $\mathbf{3,700}$. Check: $3,700 \div 4 = 925$ ✓ (remainder 0).

Day 3 – Factors, Multiples, Prime & Composite

1. Factors of 36: $\mathbf{1, 2, 3, 4, 6, 9, 12, 18, 36}$
2. Factors of 48: $\mathbf{1, 2, 3, 4, 6, 8, 12, 16, 24, 48}$
3. Factors of 60: $\mathbf{1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60}$
4. Factors of 100: $\mathbf{1, 2, 4, 5, 10, 20, 25, 50, 100}$

Prime/Composite: 17 = $\mathbf{P}$; 24 = $\mathbf{C(4 \times 6)}$; 37 = $\mathbf{P}$; 49 = $\mathbf{C(7 \times 7)}$; 51 = $\mathbf{C(3 \times 17)}$; 97 = $\mathbf{P}$

5. GCF(24, 36): Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36. Common: 1, 2, 3, 4, 6, 12. GCF = $\mathbf{12}$
6. GCF(45, 60): 45 = 1, 3, 5, 9, 15, 45. 60 = 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60. GCF = $\mathbf{15}$
7. LCM(4, 6): multiples of 4: 4, 8, 12... multiples of 6: 6, 12... LCM = $\mathbf{12}$
8. LCM(8, 12): multiples of 8: 8, 16, 24... multiples of 12: 12, 24... LCM = $\mathbf{24}$

Week 4 Answer Key – Decimals

Day 1 – Decimal Place Value

Place/Value table: 47.305 → tens, 70 | 12.486 → hundredths, 0.08 | 0.053 → tenths, 0 | 304.091 → hundredths, 0.09 | 5.738 → hundredths, 0.03

1. "eight and forty-seven thousandths" 2. "three hundred five thousandths"
3. **14.062** 4. **300.8**

5. $0.7 = \frac{7}{10}$ 6. $0.43 = \frac{43}{100}$ 7. $0.125 = \frac{125}{1000} = \frac{1}{8}$ 8. $9/100 = 0.09$

9. $0.6 = 0.60 = 0.600 \rightarrow$ **all equal**
10. $3.45 = 3.450 > 3.405 \rightarrow$ **3.45 = 3.450 > 3.405**
11. Least to greatest: **0.08, 0.081, 0.80, 0.8**
12. Greatest to least: **4.70, 4.7, 4.071, 4.07**

Day 2 – Rounding Decimals

6.382: whole=**6**, tenth=**6.4**, hundredth=**6.38**
0.547: whole=**1**, tenth=**0.5**, hundredth=**0.55**
14.905: whole=**15**, tenth=**14.9**, hundredth=**14.91** (5 rounds up)
2.999: whole=**3**, tenth=**3.0**, hundredth=**3.00**
0.0047: whole=**0**, tenth=**0.0**, hundredth=**0.00**

Number line: 0.3 is 3 tick marks from 0; 0.7 is 7 ticks; 0.15 is halfway between 0.1 and 0.2.
Halfway between 0.4 and 0.5 = **0.45** Between 3.6 and 3.7 = **3.65**

4. 8.6 oz 5. \$3.88/gal 6. 9.843 rounds to 9.8 → **faster than 9.9**
7. 2,000,000 8. $GCF(40,56)=8$ 9. $5/6 + 1/4$: LCD=12 → $10/12 + 3/12 = 13/12 = 1 \frac{1}{12}$

Challenge: Largest value that rounds to 4.6 (to nearest tenth): **4.649...** → 4.649 (since 4.650 rounds to 4.7). Smallest: **4.550** (since 4.55 rounds up to 4.6).

Day 3 – Adding & Subtracting Decimals

Remember: always add zeros and line up decimal points!

1. $4.30 + 2.85 = 7.15$ 2. $13.700 + 0.048 = 13.748$ 3. $8.40 - 3.27 = 5.13$
4. $10.000 - 4.375 = 5.625$ 5. $6.003 + 14.970 + 0.800 = 21.773$ 6. $25.06 - 18.40 = 6.66$

7. $3.4 + 0.06 = 3.46$ — **YES, this is correct.** $3.40 + 0.06 = 3.46$ ✓
8. $5 - 1.2 = 3.8$. Student said 4.2 — they subtracted 0.2 but kept 5 unchanged. Correct: $5.0 - 1.2 = 3.8$.

9. $2.49 + 1.75 + 3.08 = 7.32$
10. $3.75 + 2.80 = 6.55$. $8.5 - 6.55 = 1.95$ km
11. Carlos–Alex: $5.17 - 4.83 = 0.34$ feet. Combined: $4.83 + 4.60 + 5.17 = 14.60$ feet

Magic Square: Rows/columns/diagonals must sum to 1.5.

Summer Math Booklet - Answer Key - Week 4
Row 1: $0.7 + ? + 0.3 = 1.5 \rightarrow$ middle=0.5. Row 3: $0.2 + ? + 0.8 = 1.5 \rightarrow$ middle=0.5. Col 1: $0.7 + ? + 0.2 = 1.5 \rightarrow$ middle=0.6. Col 3: $0.3 + ? + 0.8 = 1.5 \rightarrow$ middle=0.4.

Week 5 Answer Key – Geometry & Measurement

Day 1 – Angles

Note on measuring angles with a protractor: The SVG diagrams are drawn to approximate angles. Your child's measured values may vary by $\pm 2^\circ$. Accept answers within $\pm 3^\circ$ of the listed values.

Angle 1: approx **40°** , **Acute**

Angle 2: approx **115°** , **Obtuse**

Angle 3: approx **65°** , **Acute**

Angle 4: approx **145°** , **Obtuse**

Drawing angles: Child should use protractor from the given ray. Check: ray is at stated degree from baseline.

Part C: 1. $90 - 37 = \mathbf{53^\circ}$ 2. $180 - 124 = \mathbf{56^\circ}$ 3. $360 - 90 - 115 - 82 = \mathbf{73^\circ}$
4. Angles are x and $3x$. $x + 3x = 180$. $4x = 180$. $x = 45$. Angles: **45° and 135°**

Clock Challenge: Each hour = 30° . 3:00: 3 hours = **90°** . 6:00: 6 hours = **180°** . 2:00: 2 hours = **60°**

Day 2 – Triangles & Quadrilaterals

Triangle classification: Use protractor values.

Triangle 1 (equilateral-looking): All angles $\approx 60^\circ$. Type: **Equilateral (sides); Acute (angles)**

Triangle 2 (right): $B = 90^\circ$, legs roughly equal. Angle $A \approx 45^\circ$, $C \approx 45^\circ$. Type: **Isosceles (sides); Right (angles)**

Triangle 3 (scalene): Three different angles. Measure to verify. Type: **Scalene (sides); Varies**

1. $180 - 48 - 75 = \mathbf{57^\circ}$ 2. $180 - 90 - 36 = \mathbf{54^\circ}$ 3. $360 - 85 - 110 - 95 = \mathbf{70^\circ}$
4. Parallelogram: opposite angles equal; adjacent supplementary. If one is 68° : **68° , 112° , 68° , 112°**

Quadrilateral sort: Square | Rectangle | Rhombus | Trapezoid | Parallelogram (answer for last one)

Angle Chase: AB is a straight line = 180° . The angle on the left of the lines at vertex = 42° .

x (between the two drawn lines going up) + 42° + (angle on right of right-line) = 180° (straight line).

Without the exact figure, using the typical setup:

x is vertically opposite or supplementary to 42° .

$x = 138^\circ$ (supplement of 42°)

y is the angle between lines AC and AD. $z = 180 - x - y$ relationship depends on figure layout.

Walk through with your child using the actual printed figure and protractor to verify.

Day 3 – Area & Perimeter

Shape 1 (Rectangle 14.5×8): $P = 2(14.5 + 8) = 2(22.5) = \mathbf{45 \text{ cm}}$. $A = 14.5 \times 8 = \mathbf{116 \text{ cm}^2}$

Shape 2 (Triangle base 16, height 9): $A = \frac{1}{2} \times 16 \times 9 = \mathbf{72 \text{ cm}^2}$

Shape 3 (Square 7.5): $P = 4 \times 7.5 = \mathbf{30 \text{ m}}$. $A = 7.5^2 = \mathbf{56.25 \text{ m}^2}$

Shape 4 (Parallelogram base 14, height 7): $A = 14 \times 7 = \mathbf{98 \text{ cm}^2}$

L-shape: Large rectangle $(10 \times 8) = 80$. Remove corner rectangle $(5 \times 4) = 20$. Area = $80 - 20 = \mathbf{60 \text{ ft}^2}$

OR: split into two rectangles: $(10 \times 4) + (5 \times 4) = 40 + 20 = 60 \text{ ft}^2$